

Do Recurrent RL Agents Discover a Unique Internal Clock?

Evidence Against a Canonical Temporal Code

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Keywords: temporal representation, recurrent neural networks, reinforcement learning, representational analysis, CKA

Summary

We examine how LSTM-based RL agents frame temporal information in their hidden states when no explicit clock is provided. Using CKA comparison with engineered encodings, linear probing, and PCA, we find that temporal encoding is training-dependent, non-unique across random seeds, and sensitive to temporal horizon. These findings challenge the implicit assumption that recurrent networks discover a single canonical temporal code, and suggest that the framing of time in learned representations is contingent on optimization path and task structure.

Contribution(s)

1. We show that temporal encoding in recurrent RL agents is not unique—different training runs produce different internal clocks despite identical behavior.
Context: Challenges the implicit “canonical temporal code” assumption in RL representation analysis.
2. We demonstrate that the representational geometry shifts with temporal horizon, from sinusoidal-like to scalar-like encoding.
Context: Links task structure to the learned framing of time, with implications for transfer and generalization.
3. We provide a simple, controlled methodology (CKA against engineered codes + multi-seed comparison) for testing uniqueness assumptions about learned representations.
Context: Applicable beyond time to other latent task variables.

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Abstract

1 Recurrent RL agents that must track time without an explicit clock have to invent their
 2 own internal framing of the temporal dimension. A natural assumption is that this
 3 framing is unique—that any sufficiently trained agent converges to the same canon-
 4 ical temporal code. We show this assumption is wrong: LSTM agents trained on the
 5 same periodic task construct fundamentally different internal representations of time
 6 depending on their random initialization. Using Centered Kernel Alignment (CKA)
 7 against engineered temporal encodings, linear probing, and PCA on DoorGrid, an en-
 8 vironment with periodic dynamics, we ask three questions: (Q1) Is temporal structure
 9 learned or inherent to recurrence? (Q2) Is the emergent encoding unique across training
 10 runs? (Q3) How does the encoding depend on the temporal horizon? A random-weight
 11 LSTM control shows that temporal structure is weak before training ($\text{CKA} \approx 0$, phase
 12 accuracy at chance) but emerges reliably after RL training. Multi-seed experiments
 13 ($T = 4, 5$ seeds) reveal that the learned encoding is *non-unique*: all agents achieve
 14 perfect timestep prediction ($R^2 = 1.0$) yet exhibit different CKA profiles—the highest-
 15 CKA engineered encoding varies across seeds from sinusoidal positional encoding to
 16 phase-based encoding. A controlled horizon manipulation ($T \in \{4, 8, 16, 32\}$, 5 seeds
 17 each) shows that representation similarity is *horizon-sensitive*: CKA with raw scalar en-
 18 coding increases from 0.40 ± 0.10 at $T = 4$ to 0.70 ± 0.22 at $T = 32$, while sinusoidal
 19 PE decreases from 0.63 ± 0.09 to 0.50 ± 0.08 . These findings challenge the implicit
 20 assumption that recurrent networks converge to a single canonical temporal code; the
 21 framing of time in learned representations is contingent on both initialization and task
 22 structure.

23 1 Introduction

24 RL researchers typically assume the hidden state of a recurrent network is a unified representation of
 25 the agent’s situation—including elapsed time. But when no clock signal is provided, the agent has to
 26 *construct its own framing* of the temporal dimension. This invites a question: given a well-defined
 27 POMDP, do trained agents converge to a *canonical* temporal code—a unique, reproducible internal
 28 clock? Our answer is no: different training runs produce different temporal codes despite identical
 29 behavioral performance.

30 Prior work has established that time cells emerge in RNN-based RL agents, analogous to hippocam-
 31 pal time cells (MacDonald et al., 2011; Lin et al., 2023). In NLP, the form of positional encoding—
 32 sinusoidal, learned, or relative—has been a central design choice in the Transformer architecture
 33 (Vaswani et al., 2017). Less is known about the *variability and contingency* of such representations
 34 in recurrent RL agents.

35 We address three concrete questions:

36(Q1) Does temporal structure emerge through training, or is it inherent to recurrent architectures?

37(Q2) Is the emergent encoding unique across training runs?

38(Q3) How does the encoding depend on the temporal horizon of the task?

39 We study these questions by training LSTM agents on DoorGrid, an environment with periodic
40 dynamics, and comparing their hidden states against a library of engineered temporal encodings
41 using Centered Kernel Alignment (CKA) (Kornblith et al., 2019), complemented by linear probing,
42 PCA, and single-neuron phase selectivity analysis.

43 Our central finding is that the temporal framing learned by LSTM agents is *non-unique*: agents
44 with identical task performance align differently with simple engineered temporal codes, and the
45 preferred alignment shifts with temporal horizon. This calls into question how we interpret and
46 compare recurrent representations across training runs—if there is no single canonical code, what
47 exactly are we comparing?

48 2 Related Work

49 2.1 Temporal Representations in RL

50 Lin et al. (2023) showed that time cells emerge in RNN-based RL agents, extending the neuroscience
51 observation that hippocampal neurons encode elapsed time (MacDonald et al., 2011). Hausknecht
52 and Stone (2015) introduced DRQN, showing that recurrent connections enable RL agents to handle
53 partial observability. Kapturowski et al. (2019) scaled this with R2D2 using LSTM-based agents on
54 Atari. Jaderberg et al. (2017) showed that auxiliary temporal prediction tasks improve representation
55 learning. Our work examines not just *whether* agents encode time, but whether this encoding has a
56 unique canonical form.

57 2.2 Probing Neural Representations

58 Linear probing (Alain and Bengio, 2017) and CKA (Kornblith et al., 2019) are standard tools for
59 analyzing representations. Hewitt and Manning (2019) emphasized that probe accuracy alone is
60 insufficient—one must control for selectivity. Voita and Titov (2020) studied information-theoretic
61 aspects of probing. CKA is used here as a descriptive similarity measure to a small library of
62 candidate temporal codes, not as an identification procedure for the true latent code.

63 2.3 Multiple Solutions in Neural Networks

64 The existence of multiple loss landscape basins is well documented (Garipov et al., 2018; Li et al.,
65 2018; Draxler et al., 2018). In neuroscience, Edelman and Gally (2001) formalized *degeneracy*—
66 distinct structures performing the same function. We ask a question that sits at the intersection
67 of these lines of work: do different optimization basins produce qualitatively different *temporal*
68 *representations* for the same RL task?

69 3 Methods

70 3.1 Environment

71 **DoorGrid** is an 8×8 grid with a periodic door that alternates between open and closed. At $T = 4$,
72 the door is open during phases 0–1 and closed during phases 2–3. The agent receives a one-hot
73 encoding of its (x, y) position (64-dimensional); no explicit temporal features are provided. Actions
74 are {up, down, left, right}. The reward is +1 for reaching the goal (top-right corner), −1 for stepping
75 into a hazard (a cell that is terminal when the door is closed), and 0 otherwise. Episodes terminate
76 upon reaching the goal, hitting a hazard, or exceeding a step limit. For horizon scaling, we vary

Table 1: Trained vs. untrained LSTM on DoorGrid ($T = 4$). $R^2(t)$: timestep prediction. $\text{Acc}(\phi)$: phase classification. CKA Best: highest CKA with any engineered encoding. Return: best evaluation reward (20-episode mean). Trained row shows seed 42 as representative; all 5 seeds achieve $R^2 = 1.0$ and Return = 0.920. Untrained LSTM does not participate in the RL task, so Return is not applicable.

Condition	$R^2(t)$	$\text{Acc}(\phi)$	CKA Best	MI norm	Return
Trained (seed 42)	1.000	1.000	0.740	1.000	0.920
Untrained	0.304	0.298	0.153	0.082	N/A

77 $T \in \{4, 8, 16, 32\}$; the door is open during the first half of each period and closed during the second
78 half.

79 We chose this minimal environment deliberately: the periodic temporal structure is the agent’s pri-
80 mary challenge, letting us isolate temporal encoding from complex spatial navigation.

81 3.2 Agents and Training

82 A single-layer LSTM ($d = 64$) with linear policy and value heads, trained with REINFORCE with
83 baseline (learning rate 10^{-3} , entropy coefficient 0.01, discount $\gamma = 0.99$). An MLP baseline (2-
84 layer, 64 units each) processes each observation independently, serving as a sanity check for the
85 necessity of recurrence. All experiments use 5 random seeds $\{42, 123, 456, 789, 1024\}$.

86 3.3 Analysis Pipeline

87 After training, we run 200 evaluation episodes and collect hidden state vectors $\mathbf{h}_t \in \mathbb{R}^{64}$ along with
88 timestep t and phase $\phi = t \bmod T$.

89 **Linear Probe.** Ridge regression from $\mathbf{h}_t$ to t (R^2 score) and logistic regression to ϕ (classification
90 accuracy), with shuffled-target controls ($R^2_{\text{shuffled}} = 0.020$, confirming no leakage).

91 **PCA.** PC1 variance fraction and its correlation with timestep.

92 **Mutual Information.** Normalized mutual information $I(h_d; \phi)/H(\phi)$ between each hidden dimen-
93 sion and phase, averaged over dimensions, quantifying how much phase information is distributed
94 across the representation.

95 **CKA.** Linear CKA (Kornblith et al., 2019) between $\mathbf{h}_t$ and engineered temporal encodings: raw
96 scalar (t/T), sinusoidal PE ($\sin / \cos$ of $2\pi t/T$), phase $\sin/\cos$, one-hot phase, and remaining time
97 $((T - t)/T)$. As reference baselines, CKA between engineered encodings themselves ranges from
98 0.38 to 0.54 at $T = 8$, and CKA between any encoding and shuffled hidden states is approximately
99 0.

100 **Single-Neuron Phase Selectivity.** For each hidden dimension d , we compute the mean activation
101 per phase $\bar{h}_d(\phi)$ and a selectivity index $s_d = (\max_{\phi} \bar{h}_d - \min_{\phi} \bar{h}_d) / (\max_{\phi} \bar{h}_d + |\min_{\phi} \bar{h}_d| + \epsilon)$.
102 Neurons with $s_d \approx 1$ are strongly phase-tuned. We compare the top- k phase-selective dimensions
103 across seeds.

104 4 Results

105 4.1 Temporal Structure Emerges Through Training (Q1)

106 Table 1 speaks to Q1. Trained LSTMs show strong temporal encoding ($R^2 = 1.0$, CKA = 0.74,
107 phase accuracy 1.0) while untrained LSTMs show CKA ≈ 0 and phase accuracy at chance (0.30
108 for 4 classes). The temporal structure we measure is largely training-dependent rather than a mere

Table 2: CKA (mean $\pm$ std across 5 seeds) between DoorGrid ($T = 4$) hidden states and engineered encodings. High std relative to mean indicates seed-dependent encoding. Highest-CKA mean in **bold**.

Encoding	CKA (mean $\pm$ std)
Raw scalar (t/T)	0.40 ± 0.10
Sinusoidal PE	0.63 ± 0.09
Phase sin/cos	0.28 ± 0.21
One-hot phase	0.27 ± 0.16

Table 3: Representation similarity shifts with temporal horizon in DoorGrid. CKA (mean $\pm$ std, 5 seeds) with engineered temporal encodings. [†]1 seed failed at $T = 16$ and 2 at $T = 32$; stats computed on all 5. Converged-only: $T = 16$: CKA(raw)=0.59, sin_PE=0.65; $T = 32$: CKA(raw)=0.61, sin_PE=0.54.

T	CKA(raw)	CKA(sin_PE)	PC1-t	$R^2(t)$	Return
4	0.40 ± 0.10	0.63 ± 0.09	0.44 ± 0.10	1.000	0.920
8	0.56 ± 0.14	0.58 ± 0.08	0.46 ± 0.29	1.000	0.880
16	0.64 ± 0.15	0.60 ± 0.10	0.63 ± 0.37	1.000	0.800 [†]
32	0.70 ± 0.22	0.50 ± 0.08	0.71 ± 0.28	1.000	0.640 [†]

109 artifact of random recurrent dynamics. As expected, the MLP baseline (which lacks recurrence) fails
 110 to learn the task (Return = -2.0 , $R^2 < 0.04$), confirming that temporal tracking requires recurrent
 111 state.

112 4.2 The Temporal Framing Is Not Unique (Q2)

113 All 5 seeds achieve $R^2 = 1.0$, phase accuracy = 1.0, and Return = 0.920—functionally identical
 114 performance. Yet the CKA profiles differ substantially across seeds. Of 5 seeds, 4 show high-
 115 est CKA with sinusoidal PE (0.64–0.74), while seed 456 shows highest CKA with phase sin/cos
 116 (0.62) with sinusoidal PE scoring lower (0.46). The coefficient of variation across seeds is 0.15
 117 for CKA(sin_PE) and 0.76 for CKA(phase), indicating that the encoding geometry is sensitive to
 118 random initialization. Cross-seed CKA between the agents’ own hidden states (aligned by timestep)
 119 is uniformly high (> 0.96), so the overall representational geometry is similar across seeds. The
 120 plurality we identify is not in the raw hidden-state structure but in the *alignment with specific en-*
 121 *gineered codes*: the same population geometry can be more or less similar to different candidate
 122 temporal codes depending on initialization.

123 **Single-neuron evidence.** We identify the 8 hidden dimensions with highest phase selectivity ($s_d \approx$
 124 1.0 for all seeds) as “time cells.” Across all 10 pairs of seeds, the mean overlap in top-8 time cell
 125 dimensions is 1.8/8 (mean Jaccard index = 0.14). However, with 34 of 64 dimensions showing
 126 near-maximal selectivity ($s_d > 0.9$), the hypergeometric expectation for random overlap is $8 \times$
 127 $8/34 \approx 1.88$ —making the observed overlap indistinguishable from chance. We therefore do not
 128 treat single-neuron overlap as evidence for or against representational plurality; the population-level
 129 CKA differences (above) provide the stronger signal. Because hidden-unit indices are not aligned
 130 across independently trained models, we treat the overlap analysis as descriptive (see Appendix A
 131 for per-seed details).

132 4.3 The Framing Shifts with Temporal Horizon (Q3)

133 Table 3 and Figure 1 answer Q3. At $T = 4$, mean similarity to sinusoidal PE is highest (CKA
 134 = 0.63 vs. raw scalar = 0.40). At $T = 8$, the two are roughly equal (CKA ≈ 0.57). At $T \geq 16$,

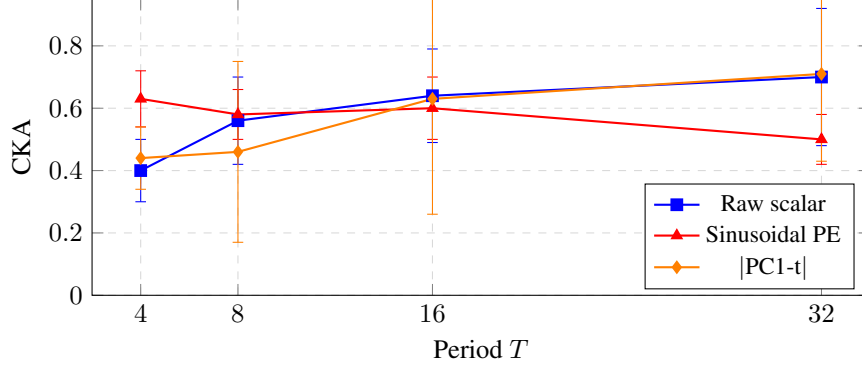


Figure 1: CKA with engineered encodings and IPC1-timestep correlation as a function of period T (mean $\pm$ std across 5 seeds). Similarity to raw scalar encoding increases at longer horizons, while similarity to sinusoidal PE decreases.

135 similarity to raw scalar exceeds sinusoidal PE. The IPC1-timestep correlation also increases with
136 period length (from 0.44 at $T = 4$ to 0.71 at $T = 32$), indicating that temporal information becomes
137 more concentrated in a single dominant dimension at longer horizons.

138 The seed-level variability persists across all period lengths. At $T = 8$, seed 42 shows highest CKA
139 with raw scalar (0.81) while seeds 123/456/789 show highest CKA with sinusoidal PE (0.48–0.58).
140 At $T = 16$, seed 1024 matches raw scalar (0.84) while seed 789 matches remaining time (0.77).
141 The framing of time is not an artifact of a single period length.

142 Note that at $T = 32$, training becomes less stable: 2 of 5 seeds fail to converge, and task return
143 drops to 0.640 (see Appendix B for failed-seed training curves). The CKA shift at long horizons
144 may partly reflect optimization difficulty rather than purely representational adaptation.

145 5 Discussion

146 LSTM-based RL agents do not converge to a single canonical temporal code. The code they learn
147 depends on their initialization, and which engineered encoding it resembles shifts with the task’s
148 temporal horizon. These findings have implications at several levels.

149 **Conceptual implication: no canonical internal clock.** The assumption that recurrent networks
150 provide a unified, well-defined temporal code is embedded in much of RL practice—for example,
151 when comparing representations across seeds or architectures, or when interpreting hidden states as
152 encoding specific task variables. Our results indicate that, at least in simple periodic environments,
153 this assumption does not hold: the framing of time is contingent on optimization path and task
154 structure. This has direct methodological implications: a study using only seed 42 would conclude
155 “LSTMs encode time via sinusoidal-like codes,” while a study using only seed 456 would conclude
156 “LSTMs encode time via phase-based codes.” Neither generalizes.

157 **Implications for interpretability.** The representational plurality we observe raises a deeper ques-
158 tion for RL interpretability: if the temporal code is not unique, what does it mean to “interpret” a
159 neuron as a time cell? The same functional role (phase discrimination) is implemented by different
160 neuronal subpopulations across seeds. This is consistent with Edelman and Gally (2001)’s concept
161 of degeneracy—structurally different elements performing the same function—and suggests that
162 single-seed interpretability analyses may be misleading.

163 **Conceptual implications: what does “representation” mean in RL?** If temporal codes are non-
164 unique, then “what the agent has learned” is not a well-defined mapping from task to internal code,
165 but one of many functionally equivalent solutions. This is Edelman’s *degeneracy* (many-to-one
166 structure-to-function mapping), not mere *redundancy* (duplicate copies of the same structure): the

solutions differ qualitatively, not just numerically. If the representational code is path-dependent, then representation analysis characterizes one among many valid solutions, not a canonical internal structure. Whether this plurality is a nuisance for interpretability or a genuine feature of learning systems remains an open question.

Scope and limitations. We do not claim a universal taxonomy of temporal coding in RL. Instead, we isolate a simple setting where temporal structure can be varied in a controlled way and representation geometry inspected directly. Three caveats are worth flagging. First, our experiments use one environment, one algorithm (REINFORCE), and one architecture (single-layer LSTM). Second, at $T = 32$, training instability (2/5 seeds fail) limits the horizon analysis. Third, CKA measures linear similarity and may miss nonlinear encoding strategies. Future work should investigate whether representational plurality persists across architectures (GRUs, attention-based models) and in more complex domains.

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225 A Single-Neuron Phase Selectivity Analysis

Table 4: Top-8 time cell dimensions and preferred phases per seed (DoorGrid, $T = 4$). Mean pairwise overlap: 1.8/8 dimensions (Jaccard = 0.14). Hypergeometric expectation given 34/64 selective dims: ≈ 1.88 .

Seed	Top-8 dims	Preferred phases
42	28, 43, 7, 44, 31, 32, 50, 39	2, 2, 0, 1, 3, 1, 3, 2
123	2, 40, 25, 49, 14, 4, 33, 36	1, 2, 3, 3, 3, 1, 3, 2
456	40, 35, 44, 39, 36, 22, 26, 43	3, 2, 0, 0, 3, 2, 3, 2
789	25, 51, 10, 15, 29, 18, 24, 38	1, 1, 1, 3, 3, 3, 3, 3
1024	43, 10, 25, 11, 39, 4, 36, 40	3, 1, 2, 3, 2, 1, 3, 2

- 226 Hidden-unit indices are not aligned across independently trained models, so low overlap in unit
 227 indices is partially expected under permutation invariance. With 34 of 64 dimensions showing
 228 high phase selectivity, the observed overlap (1.8/8) is close to the hypergeometric chance level
 229 (≈ 1.88), so this analysis does not provide evidence for or against representational uniqueness. The
 230 population-level CKA differences provide the stronger signal: seed 42 has $\text{CKA}(\sin_PE) = 0.74$
 231 while seed 456 has $\text{CKA}(\text{phase}) = 0.62$ with $\text{CKA}(\sin_PE) = 0.46$, indicating genuinely different
 232 encoding geometries.

233 B Failed Seeds at Long Horizons

- 234 At $T = 16$, seed 1024 fails to converge (best eval reward = -2.0). At $T = 32$, seeds 456 and
 235 789 fail to converge (best eval reward = -2.0). The reported CKA statistics in Table 3 include all
 236 5 seeds, as even non-converged agents contain representational structure worth characterizing. At
 237 $T = 32$, excluding the 2 failed seeds: $\text{CKA}(\text{raw}) = 0.61 \pm 0.28$ (vs. 0.70 ± 0.22 with all 5), and
 238 $\text{CKA}(\sin_PE) = 0.54 \pm 0.10$ (vs. 0.50 ± 0.08). The same pattern holds—raw scalar still beats sin
 239 PE.